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A Proof of the main theorem

We now prove the main theorem.

Proof of Theorem 1. Let $K := \min_{c \in \mathcal{C}} |\mathcal{E}_c|$ and $k := \max_{c \in \mathcal{C}} |\mathcal{V}_c|$. Also, recall that $\delta = |\hat{p}(\mathcal{A}) - p(\mathcal{A})|$ which can equivalently be written as $\delta = |p(\mathcal{B}) - \hat{p}(\mathcal{B})|$, where $\mathcal{A}, \mathcal{B}$ denote responses that are *above* and *below* threshold:

$$\mathcal{A} := \{(c, r) \in \mathcal{X} \mid \hat{p}(r \mid c) > 1/K\} \tag{4}$$

$$\mathcal{B} := \{(c, r) \in \mathcal{X} \mid \hat{p}(r \mid c) \leq 1/K\}. \tag{5}$$

Partition the hallucination and misclassification rates into above and below threshold rates:

$$\begin{aligned} \mathrm{err} &= \hat{p}(\mathcal{A} \setminus \mathcal{V}) + \hat{p}(\mathcal{B} \setminus \mathcal{V}) \\ \mathrm{err}_{\mathrm{iiv}} &= D(\mathcal{A} \setminus \mathcal{V}) + D(\mathcal{B} \cap \mathcal{V}). \end{aligned}$$

Above the threshold, misclassifications $D(\mathcal{A} \setminus \mathcal{V})$ are the sum of $D(c, r)$ only over $(c, r) \in \mathcal{A}$ such that $r \in \mathcal{E}_c$—each contributing $D(c, r) = \mu(c)/2|\mathcal{E}_c| \leq \mu(c)/2K$. But each such misclassification also contributes $\mu(c)\hat{p}(r \mid c) \geq \mu(c)/K$ to hallucinations above the threshold $\hat{p}(\mathcal{A} \setminus \mathcal{V})$. Hence,

$$\hat{p}(\mathcal{A} \setminus \mathcal{V}) \geq 2D(\mathcal{A} \setminus \mathcal{V})$$

Thus, it remains only to show that below the threshold:

$$\hat{p}(\mathcal{B} \setminus \mathcal{V}) \geq 2D(\mathcal{B} \cap \mathcal{V}) - \frac{k}{K} - \delta. \tag{6}$$

By definition, $2D(\mathcal{B} \cap \mathcal{V}) = p(\mathcal{B} \cap \mathcal{V}) = p(\mathcal{B})$. Also, there are $|\mathcal{V}_c| \leq k$ valid responses for each c, each one in $\mathcal{B}$ having $\hat{p}(r \mid c) \leq 1/K$, so $\hat{p}(\mathcal{B} \cap \mathcal{V}) \leq \sum_c \hat{p}(c)k/K = k/K$. Hence,

$$\begin{aligned} 2D(\mathcal{B} \cap \mathcal{V}) - \hat{p}(\mathcal{B} \setminus \mathcal{V}) &= p(\mathcal{B}) - \hat{p}(\mathcal{B} \setminus \mathcal{V}) \\ &= p(\mathcal{B}) - (\hat{p}(\mathcal{B}) - \hat{p}(\mathcal{B} \cap \mathcal{V})) \\ &\leq \delta + \hat{p}(\mathcal{B} \cap \mathcal{V}) \leq \delta + \frac{k}{K}. \end{aligned}$$

This is equivalent to Eq. (6), as needed. □

B Arbitrary-facts analysis

We begin by reviewing the Good-Turing (GT) estimator of missing mass (Good, 1953) and its guarantees (McAllester and Ortiz, 2003). In that setting, N iid samples $s \sim \nu^N$ are drawn from distribution ν over set $\mathcal{S}$—abstentions are not a consideration. The missing mass is the probability that a new example drawn from ν would not be in the training sample s, and the estimate GT is the fraction of training samples that occur exactly once. We first state the prior guarantees and then adapt them to our setting with abstentions. A guarantee of McAllester and Ortiz (2003) can be stated as:

Corollary 3. *(McAllester and Ortiz, 2003) Let $s \sim \nu^N$ be N iid samples from distribution ν over set $\mathcal{S}$. Let $M := \Pr_{x\sim\nu}[x \notin s]$ and* GT *be the fraction of samples that occur exactly once. For any $\gamma \in (0, 1]$:*

$$\Pr_{s\sim\nu^N}\left[\ |M - \mathrm{GT}| \leq \frac{1}{N} + 2.42\sqrt{\frac{\ln(4/\gamma)}{N}}\ \right] \geq 1 - \gamma.$$

Proof. Let $\overline{\mathrm{GT}} := \mathbb{E}[\mathrm{GT}]$ and $\overline{M} := \mathbb{E}[M]$. The corollary follows by combining concentration bounds on M and GT. First Theorem 1 of McAllester and Schapire (2000) shows:

$$\overline{\mathrm{GT}} - \overline{M} \in [0, 1/N]$$

Then, Theorems 10 and 16 (McAllester and Ortiz, 2003) imply that with probability $\leq \exp(-N\varepsilon^2)$, M will deviate from $\overline{M}$ by more than ε in either direction, together, by the union bound giving for $\varepsilon := \sqrt{\frac{\ln(4/\gamma)}{N}}$,

$$\Pr_{s\sim\nu^N}\left[\ |M - \overline{M}| \geq \sqrt{\frac{\ln(4/\gamma)}{N}}\right] \leq \frac{\gamma}{4} + \frac{\gamma}{4} = \frac{\gamma}{2}.$$

Following McAllester and Schapire (2000) (Lemma 13), McDiarmid's inequality (McDiarmid, 1989) directly implies the convergence of GT, since changing any one example can change GT by at most $2/N$. Hence,

$$\Pr_{s\sim\nu^N}\left[\ |\mathrm{GT} - \overline{\mathrm{GT}}| \geq \sqrt{\frac{2\ln(4/\gamma)}{N}}\right] \leq 2\exp\left(-\frac{2\cdot\frac{2\ln(4/\gamma)}{N}}{4/N}\right) = \frac{\gamma}{2}.$$

Combining these three displayed equations, gives, by the union bound,

$$\Pr_{s\sim\nu^N}\left[\ |\mathrm{GT} - M| \geq \frac{1}{N} + (1+\sqrt{2})\sqrt{\frac{\ln(4/\gamma)}{N}}\right] \leq \frac{\gamma}{2} + \frac{\gamma}{2} = \gamma.$$

Finally, the corollary follows from $1 + \sqrt{2} \leq 2.42$. □

We now extend this to the case of an abstention response IDK which is not counted in sr. Specifically, we say a query c is *answered* in the training data if there is a training example $(c^{(i)}, r^{(i)})$ with $c^{(i)} = c$ and $r^{(i)} \neq \mathrm{IDK}$, and *unanswered* otherwise. Let

$$\mathcal{U} := \mathcal{C} \setminus \{c^{(i)} \mid i \leq N, r^{(i)} \neq \mathrm{IDK}\}$$

denote the set of unanswered queries. Of course, by memorizing a_c for answered queries, one can achieve perfect accuracy classifying the answered queries. We extend Turing's Missing Mass (MM) estimate to abstentions as follows:

$$\mathrm{MM} := \Pr_{(c,r)\sim p}[c \in \mathcal{U} \wedge r \neq \mathrm{IDK}].$$

We similarly use Corollary 3 to show that sr is a good estimator of MM:

Lemma 1. *For all N, $\gamma \in (0, 1]$:*

$$\Pr\left[\ |\mathrm{MM} - \mathrm{sr}| \leq 4.42\sqrt{\frac{\ln(5/\gamma)}{N}}\ \right] \geq 1 - \gamma.$$